

Practical Machine Learning

PGCP-AI August 2026

Duration: 120 Hours: 60 Theory (T) + 40 Lab (L) + 20 Self-Learning (SL)

Objective: To understand and hand-on implementation of Machine Learning Algorithms for ML Applications.

Prerequisites: Good knowledge of Python Programming and Statistics

Evaluation method: Theory exam – 40%, Lab exam – 40%, Internal Assessment– 20%

List of Course material

Text Book:

Machine Learning using Python, Manaranjan Pradhan, U Dinesh Kumar, Wiley India, 1st Edition

Reference Book:

1. Introduction to Machine Learning with Python – A guide for Data Scientists by Andreasmuller, Sarah Guido.
2. Machine Learning and Deep Learning with Python, scikit-learn, and TensorFlow 2, 3rd Edition by Sebastian Raschka
3. Hands-On Machine Learning with Scikit-Learn, Keras and Tensor Flow: Concepts, Tools and Techniques to Build Intelligent Systems by Aurelien Geron
4. AI and Machine Learning for Coders: A Programmer's Guide to Artificial Intelligence by Laurence Moroney
5. Machine Learning for Big Data: Hands- On for Developer by Jaseon Bell, Wiley

Note:

- **Lab assignments are indicative; faculty shall provide adequate assignments for better practice.**

Session 1: (2T + 1SL)

Theory:

- Machine Learning in Nutshell
- Supervised and Unsupervised Learning
- ML Applications
- Evaluating ML techniques

Sessions 2 & 3: (5T + 2L + 1SL)

Theory and Lab:

- Feature engineering and Data Pre-processing: Data Preparation, Feature creation,
- Data cleaning & transformation
- Data Validation & Modelling
- Feature selection Techniques
- Dimensionality reduction

Suggested Teaching Guidelines for

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Machine Learning Algorithms:

Sessions 4 & 5: (4T + 3L + 1SL)

Theory:

- Principal Component analysis (PCA)
- t-SNE
- K means
- Distance Measure and Data Preparation – Scaling & Weighting

Lab Assignments:

- Download any dataset from kaggle.
 - (a) Form n no .of clusters according to your observation.
 - (b) Get WSS value for each cluster.
 - (c) Find best K value

Session 6: (3T + 2L + 1SL)

Theory:

- Evaluation and Profiling of Clusters
- Hierarchical Clustering
- Clustering Case Study

Lab Assignments: Implement the model hierarchical clustering on any dataset loaded from kaggle.

Session 7: (2T + 2L + 1SL)

Theory and Lab:

- DBSCAN Clustering

Sessions 8 & 9: (4T + 4L + 1SL)

Theory:

- Linear Regression
- Logistic Regression
- Polynomial Regression
- Stepwise Regression
- Ridge Regression
- Lasso Regression
- Elastic Net Regression

Lab Assignments:

Download Dataset, perform linear, Ridge, Lasso, Logistic, Polynomial regression and check for MAE, MSE, RMSE and also check F1-score and explain with conclusion.

Sessions 10 & 11: (4T + 3L + 1SL)

Theory:

- Support Vector Machines
- Basic classification principle of SVM
- Linear and Nonlinear classification (Polynomial and Radial)
- Introduction to Hyperparameter Tuning
- Grid Search and Random Search
- Cross Validation for Model Selection

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- Download Air Quality Dataset from Kaggle Predict Air Quality Index using Linear regression and classify it into five categories using SVM (i.e. Very good, good, moderate, poor, worst)
- Perform Hyperparameter Tuning using GridSearchCV and RandomizedSearchCV.
- Compare tuned and untuned model performance using Accuracy and F1-Score.

Session 12: (2T + 2L + 1SL)**Theory:**

- Discriminant Analysis (Linear and Quadratic)
- K-Nearest Neighbors Algorithm

Lab Assignments:

Load any dataset from kaggle, preprocess the data and apply KNN Algorithm.

Session 13: (3T + 2L + 1SL)**Theory:**

- Decision Trees
- Classification and Regression Trees

Lab Assignments:

Load any dataset from UCI Machines preprocess the data and apply Decision tree Algorithm.

Session 14: (2T + 2L + 1SL)**Theory:**

- Bayesian analysis and Naïve Bayes classifier
- Assigning probabilities and calculating results

Lab Assignments:

Load any dataset preprocess the data and apply Naïve Bayes Algorithm.

Sessions 15 & 16: (5T + 3L + 2SL)**Theory:**

- Concept of Model Ensembling
- Random forest, Gradient boosting Machines, Model Stacking

Lab Assignments:

Implement Random Forest and gradient boosting algorithm on any dataset.

Sessions 17 & 18: (4T + 2L + 1SL)**Theory:**

- Association rules mining
- Apriori algorithms

Lab Assignments:

Generate Association Rules for shop using Apriori.

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- Moving average, Exponential Smoothing
- Holt's Trend Methods
- Holt-Winters' Methods for seasonality

Lab Assignments:

Implement the smoothing models on the time series data.

Sessions 21 & 22: (5T + 3L + 1SL)**Theory:**

- Auto-correlation (ACF & PACF)
- Auto-regression
- Auto-regressive Models
- Moving Average Models
- ARMA & ARIMA

Lab Assignments:

Download one data set and calculate Auto correlation. Explain ARMA and ARIMA model in time series and Explain difference between them by applying on any dataset.

Session 23: (3T + 2L + 1SL)**Theory:**

- ML in Real-Time, Algorithm Performance Metrics
- ROC and AOC
- Confusion Metrix
- F1 Score
- MSE and MAE

Lab Assignments:

Calculate Confusion matrix, F1 score, MSE, RMSE, MAE scores on any model.

Load a dataset in Jupyter notebook and implement any classification algorithm and generate classification report and accuracy score.

Sessions 24 & 25: (5T + 3L + 2SL)**Theory:**

- Recommendation Systems
- Data Collection & Storage, Data Filtering
- Collaborative Filtering
- Factorization Methods
- Evaluation Metrics: Recall, Precision, RMSE, Mean Reciprocal Rank, MAP at K, NDCG

Lab Assignments:

Download Movie dataset from kaggle and implement the movie recommendation system.

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Session 26: (2T + 2L + 1SL)

Theory:

- Anomaly detection
- Point, Contextual and Collective Anomaly
- Supervised and Unsupervised anomaly detection

Lab Assignments:

Apply Anomaly Detection on dataset using isolation forest algorithm and calculate the accuracy.